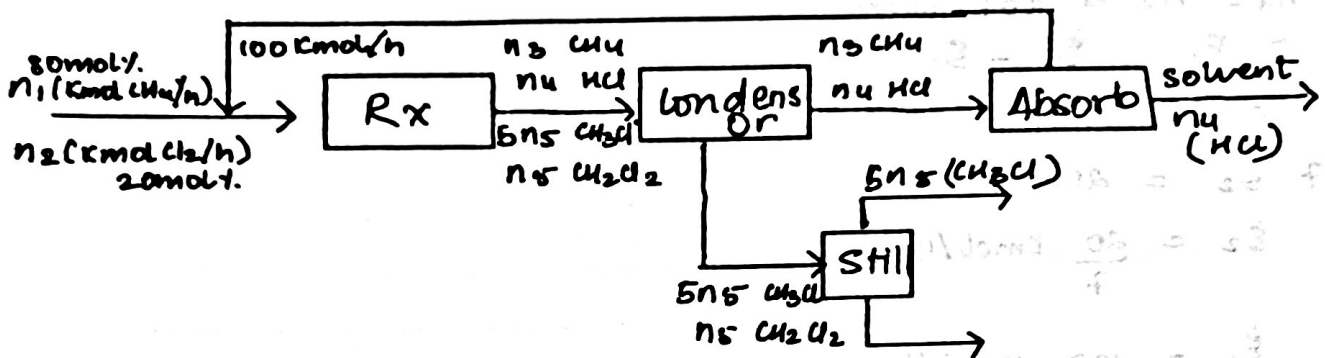
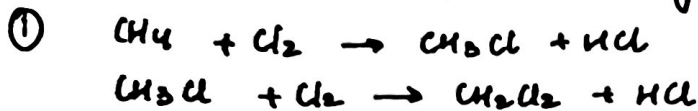


Homework 8 Raina Suamy



a) DOF overall:

$$(5+2) - (0+1+0+5) \Rightarrow 7-6=1, >0 \text{ underspecified}$$

DOF missing point

$$(5+0) - (1+1+0+2) \Rightarrow 5-4=1, >0 \text{ underspecified}$$

DOF for Reactor

$$(6+2) - (1+2+0+5) \Rightarrow 8-8=0, \text{ solvable}$$

DOF for condensor

$$(8+0) - (0+1+0+4) \Rightarrow 8-5=3 >0, \text{ underspecified}$$

DOF for Absorb:

$$(4+0) - (0+0+0+2) \Rightarrow 4-2=2 >0, \text{ underspecified}$$

DOF for still:

$$(4+0) - (0+1+0+2) \Rightarrow 4-3=1 >0, \text{ underspecified}$$

2 - 1

Reactor Species balance:

$$\text{CH}_4: \dot{n}_3 = 80 - \dot{\xi}_1$$

$$\text{HCl}: \dot{n}_4 = \dot{\xi}_1 + \dot{\xi}_2$$

$$\text{CH}_3\text{Cl}: 5\dot{n}_5 = \dot{\xi}_1 - \dot{\xi}_2$$

$$\text{CH}_2\text{Cl}_2: \dot{n}_5 = \dot{\xi}_2$$

$$\text{Cl}_2: 0 = \dot{n}_2 - \dot{\xi}_1 - \dot{\xi}_2$$

$$\dot{n}_2 = \dot{\xi}_1 + \dot{\xi}_2$$

Mixer Species balance

$$\text{CH}_4: \dot{n}_1 + \dot{n}_3 = 80$$

$$\text{Cl}_2 = \dot{n}_2$$

$$\begin{array}{r} 1) 39 \\ 2) 20 \\ 3) 30 \\ \hline 89 \\ 40 \end{array}$$

$$\frac{89}{100}$$

b. $\dot{n}_4 = \dot{n}_2 = 20 \text{ kmol/h}$

5 $\dot{\xi}_2 = \dot{\xi}_1 - \dot{\xi}_2$

6 $\dot{\xi}_2 = \dot{\xi}_1$

7 $\dot{\xi}_2 = 20 \text{ kmol/h}$

$\dot{\xi}_2 = \frac{20}{7} \text{ kmol/h}$

$\dot{\xi}_1 = \frac{120}{7} \text{ kmol/h}$

$\dot{n}_3 = 80 - \frac{120}{7} = \frac{440}{7} \text{ kmol/h}$

$\dot{n}_5 = \dot{\xi}_2 = \frac{20}{7} \text{ kmol/h}$

$\dot{n}_3 = \frac{440}{7} = 62.857 \text{ kmol/h}$

$\dot{n}_4 = 20 \text{ kmol/h}$

$\dot{n}_5 = \frac{20}{7} = 2.857 \text{ kmol/h}$

$\dot{n}_1 = 17.143 \text{ kmol/h}$

$\dot{n}_2 = 20 \text{ kmol/h}$

$X_1 \text{ CH}_4 = \frac{\dot{n}_1}{\dot{n}_1 + \dot{n}_2} = \frac{17.143}{17.143 + 20} = 0.462$

$X_2 \text{ CH}_4 = \frac{\dot{n}_3}{\dot{n}_3 + \dot{n}_4 + 5\dot{n}_5 + \dot{n}_5} = \frac{62.857}{99.999} = 0.629$

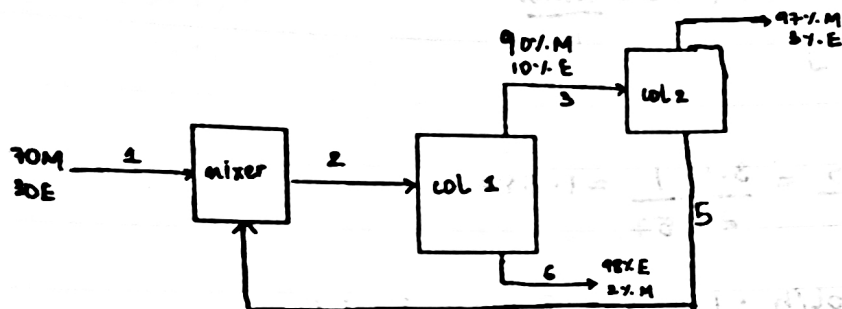
$X_2 \text{ H}_2\text{O} = \frac{\dot{n}_4}{\dot{n}_3 + \dot{n}_4 + 5\dot{n}_5 + \dot{n}_5} = \frac{20}{99.999} = 0.200$

$X_2 \text{ CH}_3\text{OH} = \frac{5\dot{n}_5}{\dot{n}_3 + \dot{n}_4 + \dot{n}_5 + 5\dot{n}_5} = \frac{14.285}{99.999} = 0.143$

$X_2 \text{ CH}_2\text{O} = \frac{\dot{n}_5}{\dot{n}_3 + \dot{n}_4 + \dot{n}_5 + 5\dot{n}_5} = \frac{2.857}{99.999} = 0.0286$

C.)? -10
↓
on pg 3

2. E for ethane and M from methane



$$70 + X_{M5}(100) = X_{M2}\dot{n}_2$$

$$\text{Mixer: } 30 + X_{E5}(100) = X_{E2}\dot{n}_2$$

$$X_{M2}\dot{n}_2 = 0.9\dot{n}_3 + 0.02\dot{n}_6$$

$$\text{Column 1: } X_{E2}\dot{n}_2 = 0.1\dot{n}_3 + 0.98\dot{n}_6$$

$$0.90\dot{n}_3 = 0.97\dot{n}_4 + X_{M5}(100)$$

$$\text{Column 2: } 0.10\dot{n}_3 = 0.03\dot{n}_4 + X_{E5}(100)$$

$$X_{M5} + X_{E5} = 1 \quad \text{and} \quad X_{M2} + X_{E2} = 1$$

We can first group all 3 units into a single system and derive two systems in two unknowns

$$70 = 0.97\dot{n}_4 + 0.02\dot{n}_6$$

$$30 = 0.03\dot{n}_4 + 0.98\dot{n}_6 \quad \text{--- (2)}$$

$$\dot{n}_4 = 71.6 \text{ gmol/min}$$

$$\dot{n}_6 = 28.4 \text{ gmol/min}$$

Total mole balance around column 2,

$$\dot{n}_3 = \dot{n}_4 + \dot{n}_5 = 71.6 + 100 = 171.6$$

Using column 2 balance equations

$$X_{E5} = 0.15 \quad \text{and} \quad X_{M5} = 0.85$$

$$X_{M2}\dot{n}_2 = 0.9(171.6) + 0.02(28.4) = 155 \text{ gmol/min}$$

$$X_{E2}\dot{n}_2 = 0.1(171.6) + 0.98(28.4) = 45 \text{ gmol/min}$$

$$\dot{n}_2 = 200 \text{ gmol/min}$$

$$X_{M2} = 0.775$$

$$X_{E2} = 0.225$$

20
30
Fractional recoveries?

$$c) \quad 5 \dot{n}_5 = 1000 \text{ kg/h} \cdot \frac{1 \text{ kmol}}{50.49 \text{ kg}} = 19.806 \frac{\text{kmol}}{\text{h}}$$

$$\dot{n}_5 = 3.961 \text{ kmol/h}$$

$$SF = \text{Scale Factor} = \frac{\dot{n}_{5, \text{new}}}{\dot{n}_5} = \frac{3.961}{2.857} = 1.386$$

$$\dot{n}_{1, \text{new}} = \dot{n}_1 \cdot SF = 17.143 \text{ kmol/h} \cdot 1.386 = 23.760 \text{ kmol/h}$$

$$\dot{n}_{2, \text{new}} = \dot{n}_2 \cdot SF = 20 \text{ kmol/h} \cdot 1.386 = 27.72 \text{ kmol/h}$$

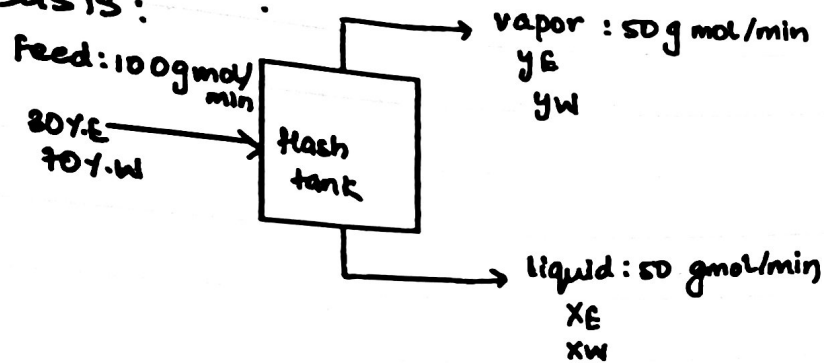
$$\dot{n}_{3, \text{new}} = \dot{n}_3 \cdot SF = 62.857 \text{ kmol/h} \cdot 1.386 = 87.120 \text{ kmol/hr}$$

$$x_{1, \text{CH}_4} = \left(\frac{\dot{n}_1}{\dot{n}_1 + \dot{n}_2} \right)_{\text{new}} = \frac{23.760 \text{ kmol/hr}}{51.480 \text{ kmol/h}} = 0.462$$

$$x_{1, \text{Cl}_2} = 1 - x_{1, \text{CH}_4} = 0.538$$

3.

Basis:



$$0.3(100) = y_E(50) + x_E(50)$$

$$0.7(100) = y_W(50) + x_W(50)$$

$$1 = y_E + y_W$$

$$1 = x_E + x_W$$

Table 13.11

@ 82.7°C

$$x_E = 0.2337$$

$$y_E = 0.5445$$

@ 84.1°C

$$x_E = 0.1661$$

$$y_E = 0.5089$$

@ 85.3°C

$$x_E = 0.1238$$

$$y_E = 0.4704$$

x_E	y_E	
$(50)(0.2337) + (50)(0.5445)$	$= 38.91$	$\neq 30$
$(50)(0.1661) + (50)(0.5089)$	$= 33.75$	$\neq 30$
$(50)(0.1238) + (50)(0.4704)$	$= 29.71$	$= \frac{30 \text{ g mol E}}{\text{min}}$

85°C will vaporize 50% of 100 g/mol and produce a vapor of 47 mol% E and liquid of 12.4 mol% E

$$\frac{30}{30}$$

Fractional Recoveries

$$f_{REV} = \frac{y_E \dot{n}_V}{x_E \dot{n}_L} = \frac{(0.47)(50)}{(0.124)(100)} = 0.78$$

$$x_W = 1 - 0.124 = 0.876$$

$$f_{RWL} = \frac{x_W \dot{n}_L}{z_W \dot{n}_F} = \frac{(0.876)(50)}{(0.7)(100)} = 0.626$$

$$x_{EW} = \frac{y_E x_W}{x_E y_W} = \frac{(0.47)(0.876)}{(0.124)(0.53)} = 6.26$$

✓